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Teaching to Support Students with Diverse Academic Needs

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Abstract

Policy and practice communities increasingly are emphasizing conceptual, cognitively demanding, and "ambitious" instruction. Within this context, we examine whether such practices serve the needs of students with specialized academic needs. Across upper-elementary classrooms in four districts, we find that exposure to "ambitious" mathematics practices is more strongly associated with test score gains of English language learners (ELs) compared to those of their peers in general education classrooms; furthermore, this teaching practice is associated with the math self-efficacy of students with individualized education programs (IEP), and the self-reported behavior of general education students. We also find links between teachers' emotional support and students' self-efficacy and engagement, with the strongest relationships for students identified both as an EL and with an IEP.

Keywords: instruction, cognitive demand, social-emotional outcomes, English language learners, special education

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Introduction

What makes for "good" or "effective" teaching, and for whom? For decades, scholars have weighed this question, using theory and empirical evidence to identify characteristics of high-quality teachers and high-quality instructional practices (e.g., Cohen, 2011; Fenstermacher & Richardson, 2005; Lampert, 2001; Porter & Brophy, 1988). Across all classrooms, grade levels, and content areas, "good" or "effective" teaching is thought to focus not only on instructional supports in distinct content areas but also on the social and behavioral needs of students, including engaging in positive interactions with students, developing students' self-regulatory and self-management behaviors, and creating an environment conducive to learning (Pianta & Hamre, 2009).

In mathematics, scholars have debated in particular the merits of procedurally based versus conceptually based teaching (e.g., Hiebert & Lefevre, 1986), with the latter largely emerging as the primary goal of mathematics reforms from the 1990s to today. Conceptual, cognitively demanding, or "ambitious" instructional practices (Cobb & Jackson, 2011) are the foundation of the Process Standards from the National Council of Teachers of Mathematics ([NCTM], 1989, 2014), the Standards of Mathematical Practice under the Common Core State Standards (National Governors Association, 2010), and the National Mathematics Advisory Panel (2008). These panels recommend, for example, providing explanations for mathematical phenomena (Fuchs, Fuchs, Hamlett, et al., 1997), linking between mathematical representations (Ball, Hill, & Bass, 2005), comparing multiple solution strategies (Rittle-Johnson & Star, 2007), and examining patterns to make generalizations (Warren & Cooper, 2008), all of which are thought to improve students' understanding of mathematical concepts.[1]

Within this context and the growing emphasis on conceptual and ambitious teaching, we consider a critical concern raised by many practitioners and policymakers (Coleman & Goldenberg, 2012; National Academies Press, 2017): *Are instructional practices aligned to conceptually based academic standards serving the academic, social, and behavioral skills of* all *students?* Namely, *are these practices benefiting students with specialized academic needs, including English language learners (ELs) or students with an individualized education program (IEP) who receive special education services?* Whereas instructional paradigms for general education students, particularly in mathematics, tend to emphasize constructivist perspectives and conceptually based teaching (Brophy, 2002; Cobb, 1994; Confrey & Kazak, 2006), views on instruction for students identified as ELs or those with an IEP are more mixed. Some emphasize the importance of conceptually based activities aligned to constructivism (Faggella-Luby et al., 2016; Huerta et al., 2016), while others advocate for teaching grounded in direct instruction that uses explicit language and instructional techniques in order to maximize ELs' or students with an IEP's understanding of the content (Doabler, Nelson, & Clarke, 2016; Spooner, Root, Saunders, & Browder, 2019).

Understanding whether and how dimensions of instruction differentially impact the outcomes of students with specialized academic needs versus general education students has important implications for practice, policy, and equity – particularly when students with and without specialized needs learn together in general education classrooms. Logistical constraints often mean that schools rely on a single instrument to observe instruction and provide feedback to teachers (Hill & Grossman, 2013). Yet, this approach would not be appropriate if the teaching practices included in each instrument differentially influenced the outcomes of different groups of students. From an equity perspective, we would not want to withhold from a subset of students

the sorts of ambitious and cognitively demanding instructional practices that are emphasized in many observation instruments and aligned to rigorous content standards. At the same time, it is essential – and mandatory under the Individuals with Disabilities Education Improvement Act (IDEA, 2004) – that students receive instruction that is the best fit for their own learning processes.

To address these questions and concerns, we examine whether exposure to a set of teaching practices emphasized by 21st-Century content standards and captured on established teacher observation instruments are equally impactful at raising test scores and self-reported social and behavioral skills for students identified as ELs, students with an IEP, or general education students.

Literature Review

Bridging Theory and Evidence on "Good" or "Effective" Teaching

An increased policy focus on teachers and their instructional practice is due in large part to accumulating and convincing evidence on the importance of individual teachers to student outcomes. Research, particularly in the field of economics of education, has identified teachers as the most important in-school resource we can provide to students, capable of producing large gains in students' academic performance (Hanushek & Rivkin, 2010) and a range of "non-cognitive", social, or behavioral skills including students' interpersonal relationships and self-management (Jennings & DiPrete, 2010; Ruzek et al., 2015). Experimental evidence in which teachers were randomly assigned to classes indicates that a one standard deviation (SD) increase in teacher effectiveness results in average improvements in students' mathematics and reading test scores between 0.1 and 0.2 SD (Kane, McCaffrey, Miller, & Staiger, 2013; Nye, Konstantopoulos, & Hedges, 2004). In other words, having a teacher at the 86th percentile of

effectiveness moves the median student up to roughly the 55th percentile in the distribution of test-score performance. Teacher effects on other student outcomes, including students' self-reported self-efficacy, grit, and behavior are similar in magnitude, and teacher effects on students' engagement and happiness in class are even larger (roughly 0.3 SD; Blazar, 2018; Kraft, 2019). These results generally are consistent with theory indicating that "good" or "effective" teachers not only impact student learning (proxied by test scores) but also improve students' social and behavioral development (Pianta & Hamre, 2009).

Additional evidence linking scores from teacher observation instruments to gains in student outcomes provides further support for theories on "good" or "effective" teaching. Protocols used in research and practice settings include the Classroom Assessment Scoring System (CLASS; Pianta, La Paro, & Hamre, 2008), Framework for Teaching (FFT; Danielson, 2011), Mathematical Quality of Instruction (MQI; Learning Mathematics for Teaching Project, 2011), Reformed Teaching Observation Protocol (RTOP; Sawada et al., 2000), and UTeach Observation Protocol (UTOP; Walkington et al., 2012). Several of these instruments were designed explicitly to capture constructivist and ambitious teaching practices. The CLASS and the MQI are the two tools used in this study. Research projects using these instruments indicate that teachers appear to be successful at raising test scores due to a combination of strong behavior management skills (Kane & Staiger, 2012) and error-free, cognitively demanding practices in discrete content areas (for mathematics, see Blazar, 2015; Sawada et al., 2002; Walkington & Marder, 2018). Comparatively, gains in students' self-reported engagement and self-efficacy are strongly predicted by teachers' interpersonal relationships with students (Blazar & Kraft, 2017; Downer, Rimm-Kaufman, & Pianta, 2007).

Teaching to Targeted Populations

An area of tension when conceptualizing, assessing, and aiming to improve instructional effectiveness in general education classrooms concerns the specialized academic needs of targeted student populations, including ELs or students with an IEP who receive special education services. Is "good" teaching always good teaching, or is it dependent on the students with whom an individual teacher works? In other words, are the "good" teaching strategies defined by theorists and reformers, which appear to improve outcomes for general education students, equally effective for ELs and students with an IEP who also learn in general education classrooms?

English Language Learners

The education production function (Todd & Wolpin, 2003) literature highlights the need for general education teachers to develop skills specific to EL's instructional needs (Loeb et al., 2014; Master et al., 2016). Unsurprisingly, much of the literature related to ELs, including several experiments, focuses on practices that support their acquisition of language and literacy skills (for a review see Janzen, 2008; see also Carlo et al. 2009; Slavin et al., 2011). While fewer studies have examined outcomes for ELs outside of reading, those few that have identify positive effects of teaching academic vocabulary and integrating intentional written and oral language instruction into science and mathematics content areas (Janzen, 2008).

With the advent of the Common Core (National Governors Association, 2010) and a broad push to integrate conceptual understanding into general education classrooms, there is some concern regarding the benefit to ELs of engaging in this work without first developing a strong foundation in written and oral language (Coleman & Goldenberg, 2012). Some have pushed for scaffolded instruction that enables ELs to engage in complex, language-intensive

tasks in mathematics and other content areas (Doabler et al., 2016; Schleppegrell, 2007; Walqui, 2006). Evidence suggests that this approach may be effective, though research to date is limited and has focused more on science instruction than mathematics (e.g., Faggella-Luby et al., 2016; Huerta et al., 2016).

Students who Receive Special Education Services

Research also suggests that teaching students with IEPs who receive special education services requires unique skills. Broadly, our review identifies the importance of explicit and direct instruction (Powell & Fuchs, 2015; Spooner et al., 2019; Witzel, Mercer, & Miller, 2003), though evidence-based best practices can depend in part on the specific disabilities that students have (Schalock et al., 2010). For students with specific learning disabilities – who make up the largest share of students with IEPs in our sample and in the U.S. population more broadly (U.S. Department of Education, 2018) – repeated reading can be impactful (Ellis & Graves, 1990; Wexler et al., 2010). In mathematics, students with learning disabilities also benefit from corrective feedback, frequent review, organizational supports, and explicit instruction (Gersten & Baker, 2000; Gersten et al., 2009; Satsangi, Hammer, & Hogan, 2018). Students with math-specific learning disabilities can benefit from conceptual-based teaching strategies (Myers, Wang, Brownell, & Gagnon, 2015) and use of schematics such as Concrete Representational Abstracts (CRA; Butler et al., 2003; Jitendra et al., 2002). While debates about the relative benefit of procedurally based versus conceptually based teaching for students with math-specific learning disabilities is ongoing (Agrawal & Morin, 2016), some argue that both types are essential for improving math achievement for this group (Schneider, Rittle-Johnson, & Star, 2011).

Research on evidence-based best practices for teaching students with special education needs often have disaggregated findings by additional disability groups, including intellectual disabilities (e.g., Martinez & Pellegrini, 2010; Göransson, Hellblom-Thibblin, & Axdorph, 2016), autism spectrum disorder (e.g., Knight & Sartini, 2015), and emotional or behavioral disabilities (e.g., Lambert, Cartledge, Heward, & Lo, 2006; Simonsen et al., 2008).[2] However, policy research focusing on equity issues also has considered outcomes of students with an IEP as a group (e.g., King-Sears, 2008), in part because school funding provided to local education agencies is a function of the learning of students with IEPs as a block (IDEA, 2004).

One policy relevant subgroup is students identified both as an EL and with an IEP, who face distinct education challenges because of the interaction between overcoming a language barrier as well as navigating their particular academic, emotional, or physical challenges (Utley, Obiakor, & Bakken, 2011). To date, the literature on teaching ELs who also are eligible for special education services is very limited (Cheatham & Barnett, 2017).

Directions for Research

Our study builds on prior work by: (a) examining differential relationships between teaching practices and student outcomes for ELs, students with an IEP, and students without specialized academic needs who learn together in general education classrooms; (b) examining teaching practices that best support outcomes of ELs who also have an IEP; and (c) specifying models that aim to limit selection or "omitted variables" bias, which has been identified as a key limitation of prior work (Alexander, Ayres, & Smith, 2015). Extant studies that are able to support causal inferences often look at instruction in laboratory settings or for short periods of time rather than in students' own classrooms over the course of a full school year. Like others in

the literature base, our study has limitations with regard to drawing causal conclusions, but benefits from data collected under real-world conditions.

Research Design

Sample

To examine relationships among teachers, teaching, and student outcomes, and how they differ across student groups, we leveraged a unique dataset from the National Center for Teacher Effectiveness (NCTE) collected over a three-year period (2010-11 through 2012-13). The full project sample includes fourth- and fifth-grade teachers (n = 310) and their students (n = 10,575) in four urban school districts on the east coast of the United States.[3] Participating teachers were generalists who taught all subject areas, which provided an opportunity to estimate the contribution of individual teachers to student outcomes that was not confounded with the effect of another teacher with whom a student engaged in the same year. Teacher-student links were verified for all study participants based on class rosters provided by teachers. Each district provided indicators for whether or not each student was identified as an EL (based on district-specific assessments) or had an IEP and was eligible to receive special education services.

As we describe in more detail in the Analysis section below, we limited our analytic sample to teachers who participated in the NCTE project for more than one year (n = 196). This approach allowed us to use measures of teaching quality captured in years other than those in which we measured student outcomes, and addressed concerns that observation scores may be biased due to the composition of students in the classroom (Campbell & Ronfeldt, 2018; Steinberg & Garrett, 2016; Whitehurst, Chingos, & Lindquist, 2015). In a robustness test, we further limited the sample to classes/teacher-years with at least five students identified as an EL

or with an IEP (n = 135) in order to examine whether results were sensitive to the makeup of students in the classroom.

We present descriptive statistics on these teachers and their students in Table 1. When we compare these samples, we find that there are no statistically significant differences in observable characteristics (p = 0.504 comparing the sample of teachers with out-of-year observation scores to the full project sample; and 0.643 comparing the sample that further restricts to teachers with five or more students with specialized academic needs to the full project sample). All three samples reflect national patterns where the vast majority of elementary teachers are white females who earned their teaching credential through traditional certification programs. Students in our samples look similar to students in many urban districts in the United States (Council of the Great City Schools, 2013). There is an even split between males and females, roughly 40% are African-American, between 22% and 29% are Hispanic, between 20% and 27% are white, and over 60% are eligible for free or reduced-price lunch. Between 17% and 27% of students were identified as ELs, between 8% and 11% had an IEP, and between 2% and 4% were identified as both an EL and having an IEP (these categories are mutually exclusive). While we do see several statistically significant differences in the composition of students between the full project sample and our two analytic sub-samples, several of these differences are by design. In particular, our sample of teachers with five or more non-general education students ensures that our dataset has a higher percentage of ELs and students with an IEP.

Data

Measures of Teaching Practice

Each year teachers videotaped three mathematics lessons, which trained raters scored on the CLASS (Pianta et al., 2008) and MQI (Learning Mathematics for Teaching Project, 2011)

observation instruments.[4] Together these instruments capture four unique teaching constructs (see Blazar, Braslow, Charalambous, & Hill, 2017 for factor analyses; Appendix Table 1 for item descriptions; and Table 2 for descriptive statistics). From the MQI, *Ambitious Mathematics Instruction* (10 items) identifies the complexity of the tasks that teachers provide to their students and their interactions around that content, including use of and explicit linking between multiple mathematical representations (e.g., numeric fraction and fraction pie), and discussion of and linking between multiple solution strategies to solve a single problem. *Mathematical Errors* (3 items) identifies any mathematical errors or imprecisions that the teachers introduce into the lesson that go uncorrected. From the CLASS, *Emotional Support* (3 items) focuses on teachers' interactions with students, including creating a positive classroom climate, and teachers' sensitivity to and respect for student ideas and perspectives. Finally, *Classroom Organization* (3 items) captures teachers' behavior management skill, and the productivity and pacing of the lesson.[5]

We used observation and scoring protocols articulated by instrument developers. For the MQI, two certified and trained raters scored each item on a scale from 1 (not present) to 3 (present and sustained) for each 7.5-minute segment of a given lesson. For the CLASS, one certified and trained rater scored each item on a scale from 1 (low) to 7 (high) for each 15-minute segment of a given lesson. We calculated teacher-level scores for each of the four constructs by first averaging scores for each item across raters (for the MQI only), then across segments, lessons, and items within the dimension.

These teacher-level scores adequately capture the quality of teachers' instruction with adjusted intra-class correlations of 0.74, 0.56, 0.53, and 0.63 for *Ambitious Mathematics Instruction, Mathematical Errors, Emotional Support,* and *Classroom Organization,*

respectively.[6] Though some of these estimates are lower than conventionally accepted levels (0.7), they are consistent with those generated from similar studies (Kane & Staiger, 2012) and do not appear to attenuate to zero estimates of the relationship between observation scores and student outcomes in other studies using these same data (Blazar, 2015; Blazar & Kraft, 2017). Attenuation due to measurement error is a primary concern with lower ICCs (Spearman, 1904). However, higher ICCs may be needed for other studies focused specifically on high-stakes decision-making. We standardized scores within the full sample of teachers to have a mean of zero and a standard deviation of one. See Table 3 for descriptive statistics on the teacher observation measures.

Student Outcomes

District administrative records allowed us to capture students' high-stakes, end-of-year state test scores in mathematics and reading. Test scores were standardized to have a mean of zero and standard deviation of one within district, subject, grade, and year using the full population of students. Lynch, Chin, and Blazar (2017) coded these state tests for their cognitive demand, as proxied by item format (i.e., multiple choice, short response, open-ended response) and alignment with the MQI instrument's emphasis on student engagement in meaning making. In two of the districts that came from the same state, the test was closely aligned to the sorts of ambitious and cognitively demanding mathematics tasks (e.g., open-ended items that asked students to look for patterns) emphasized in many content standards and by the MQI. A third district's assessment included a combination of cognitively demanding and procedural items, while the fourth district's assessment focused almost exclusively on solving basic procedures.

We also considered the relationship between teaching practices and student outcomes beyond test scores, recognizing that test scores capture only a subset of the skills that students

need when they leave school (Deming, 2017). Students' social and behavioral skills were captured on a self-reported survey administered by NCTE staff at the end of each school year. Items were adapted from other large-scale surveys/projects including Tripod, Measures of Effective Teaching (MET), National Assessment of Educational Progress (NAEP), and Trends in International Mathematics and Science Study (TIMSS). Students rated all 18 items on a five-point Likert scale where 1 = Totally Untrue and 5 = Totally True. Based on theory and exploratory factor analyses (see Blazar & Kraft, 2017 for these analyses; Appendix Table 2 for item text; and Table 3 for descriptive statistics), we identified three constructs from these data: *Self-Efficacy in Math* (10 items; internal consistency reliability [α] = 0.76), *Engagement and Happiness in Class* (5 items, α = 0.82), and *(Good) Behavior in Class* (3 items; α = 0.74). Student reports of versions of the first two constructs and teacher reports of student behavior have been linked to labor market outcomes, even controlling for cognitive ability (Lyubomirsky, King, & Diener, 2005; Mueller & Plug, 2006), lending predictive validity to these metrics. We created final scales by reverse coding items with negative valence, averaging student responses across all available items, and then standardizing to mean of zero and standard deviation of one.[7] Standardization occurred within school year, given slight differences in the wording of a handful of items across years, but across grades, given that the same survey instrument was administered to fourth and fifth graders. Items comprising the *Engagement and Happiness in Class* construct were added to the survey in year two of the study.

Analysis

We estimated the contribution of teachers and specific teaching practices to an array of student outcomes using a standard model of student production (Todd & Wolpin, 2003) for student *i* in school *s* with teacher *j* at time *t*:

$$OUTCOME_{isjt} = \alpha f(A_{it-1}) + \pi X_{it} + \varphi \overline{X}^{c}_{it} + \nu_s +$$

$$\beta\left(\sum_{n=1}^{4} Academic\,Group_i * \widehat{OBSERVATION}_{j,-t}\right) + \varepsilon_{isjt} \quad (1)$$

We predicted student-level outcomes as a function of prior academic performance (A_{it-1})[8] and a range of other observable student- (X_{it}) and class-level ($\overline{X}^{c}_{it}$) characteristics. We included school fixed effects, ν_s, to account for the non-random sorting of students and teachers across schools, which has become standard practice in the estimation of teacher effects in the academic literature (Hanushek & Rivkin, 2010). The school fixed effects also account for the fact that math achievement tests differed across districts. Finally, we included a vector of observable teaching practices captured on the MQI and CLASS instruments, $\widehat{OBSERVATION}_{j,-t}$, interacted with four mutually exclusive categories of students with diverse academic needs: (1) general education students, (2) ELs, (3) students with an IEP, and (4) students identified both as an EL and having an IEP. Our parameters of interest in the vector β describe the relationship between teaching practices and student outcomes for each group of students. We tested the hypothesis that each of these parameters was statistically significantly different from zero, and whether the relationship between one teaching practice and student outcomes differed across student groups.

Equation (1) seeks to address two primary threats to internal validity: non-random sorting of teachers to students, and non-random sorting of teaching practice to teachers. For the first concern, we relied on past experimental research showing convincing evidence that controlling for students' prior test scores accounts for sorting bias when current-year test scores are the outcome of interest (Kane et al., 2013). Evidence from a subset of the NCTE data in which teachers were randomly assigned to class rosters within schools also indicates that sorting bias can be eliminated when predicting students' *Behavior in Class* by controlling for prior test scores

(Blazar, 2018).[9] We adopted this approach in the model outlined above. Bias in teacher effects on the two other outcomes – *Self-Efficacy in Math* and *Engagement and Happiness in Class* – was reduced with a similar set of controls, but not eliminated entirely. In light of this validation work, we interpret our estimates of the relationship between teaching practices and the latter two student outcomes with caution.

The second concern for identifying differential teaching effects is the endogeneity of observed classroom quality. In other words, specific teaching practices were not randomly assigned to teachers. Our analytic approach addressed this concern in two ways. First, we conditioned estimates of the relationship between one dimension of teaching practice and student outcomes on the three other dimensions. This method allowed us to parse the effect of one teaching practice from another. While the set of teaching practices captured on the MQI and CLASS is not exhaustive, together these instruments cover a wide range of content-specific and general classroom practices. Second, similar to Kane et al. (2011) we only included lessons captured in years other than those in which student outcomes were measured, denoted by $-t$ in the subscript of $\widehat{OBSERVATION}_{j,-t}$. Limiting the lessons in this way helps ensure that unmeasured characteristics of instruction are uncorrelated with student outcomes.[10] We used predicted, shrunken observation scores to account for the fact that teachers contributed different numbers of lessons to the project, and fewer lessons could lead to greater measurement error in these scores (Hill, Charalambous, & Kraft, 2012).[11]

Results

The main results are presented in Tables 4 and 5, which show point estimates of the relationship between the four dimensions of teaching practice and the four student outcomes (former table), and *p*-values for differences between these relationships across student groups

(latter table). We hypothesized that different compositions of students in the classroom may lead to different results, which we speculate on in instances where estimates differ between our two analysis samples: the larger sample of teachers with out-of-year observation scores (column 1 in each table), and the smaller sample of teachers with out-of-year observation scores and five or more students with specialized academic needs (column 2 in each table). By and large, though, results are similar between the two samples.[12] For both, sample sizes are smaller in models predicting students' *Engagement and Happiness in Class* relative to others given that this construct only was available in two of the three years of the study that generated the data used in this paper.

In Table 4, all estimates are presented as standardized effect sizes and can be interpreted as the SD increase in student outcomes caused by or associated with a 1 SD increase in teaching practice, controlling for other teaching practices, student and class characteristics, and school fixed effects. We remind readers that, based on past validation exercises, causal inferences seem warranted when predicting students' math test scores and *Behavior in Class* but not necessarily for the other two student survey measures (Blazar, 2018). All models cluster standard errors at the class level to account for the nested structure of the data.

Detecting differential relationships across groups (Table 5) generally requires more statistical power than detecting whether each relationship differs from zero (Table 4). Therefore, in Table 5 we bold *p*-values at the conventional level of statistical significance (0.05) but also provide estimates to three decimal places to allow for readers' own interpretations.

Beginning our interpretation first with *Ambitious Mathematics Instruction*, we find exposure to teaching practice focused around cognitively demanding mathematics activities (e.g. providing mathematical explanations, linking between mathematical representations and solution

strategies) results in improvements in mathematics test scores for several groups of students (see Table 4). For ELs, exposure to teachers who score 1 SD above the mean in this dimension of teaching results in a 0.06 to 0.07 SD increase in mathematics test scores, controlling for the three other teaching practices and a range of student, class, and school characteristics. Estimates are slightly larger for students identified as both ELs and with an IEP (0.09 to 0.10 SD). Comparatively, we do not see relationships between *Ambitious Mathematics Instruction* and mathematics test scores that are statistically significantly different from zero for general education students or students with an IEP only. In Table 5, *p*-values indicate that, in fact, the relationship between *Ambitious Mathematics Instruction* and math test scores is stronger for ELs only and ELs who also have an IEP relative to students with an IEP only ($p = 0.04$ to 0.10). We also see that the relationship between *Ambitious Mathematics Instruction* and mathematics test scores is stronger for ELs than for general education students when we relax our threshold for statistical significance slightly ($p = 0.06$ to 0.08). The substantive difference in magnitudes of relationships between *Ambitious Mathematics Instruction* and math test scores for ELs (0.06 to 0.07 SD) versus general education students (0.02 SD) also supports this conclusion (see Table 4).

Considering other student outcomes, we see a positive relationship between teachers' *Ambitious Mathematics Instruction* and *Self-Efficacy in Math* for students with an IEP (0.11 SD), and between this dimension of teaching practice and students' *Behavior in Class* for both the general education population (0.06 SD) and students with an IEP (0.10 SD; see Table 4). These relationships only show up as statistically significantly different from zero in the sample of teachers with five or more students with specialized academic needs, suggesting that the composition of students in the classroom may serve as a mediator. Yet, respective point estimates in the larger sample are similar, and 95% confidence intervals around estimates

between the two samples overlap. It does not appear that these relationships are any stronger for one group than another (see Table 5).

Moving next to *Mathematical Errors*, we find that having a teacher who makes more errors in their instruction (i.e., higher scores on this dimension) results in lower mathematics test scores for general education students (-0.04 to -0.05 SD; see Table 4). The presence of *Mathematical Errors* appears to be more detrimental for general education students than for students identified both as an EL and having an IEP (p = 0.01 to 0.07; see Table 5). Oddly, we find that more errors made in teachers' mathematics instruction is positively related to most students' *Engagement and Happiness in Class*, and to the *Behavior in Class* of students with an IEP only. Understanding the mechanisms driving these counterintuitive relationships is beyond the scope of this paper. However, they are consistent with past work showing a tradeoff between teachers' contributions to students' math test scores versus student-reported *Engagement and Happiness in Class* (Blazar, 2018). More pertinent to the questions addressed in this paper, we do not find differential relationships between teachers' *Mathematical Errors* and students' *Engagement and Happiness in Class* across student subgroups. We do see that the relationship between teachers' *Mathematical Errors* and students' *Behavior in Class* is stronger (i.e., more positive) for students with an IEP compared to general education students (p = 0.02 to 0.03) and to students identified as both an EL and with an IEP (p = 0.02; see Table 5).

Consistent with theory, we find moderate to strong positive relationships between teachers' *Emotional Support* for students and the two student outcome measures that most closely align with this construct: student-reported *Self-Efficacy in Math* and *Engagement and Happiness in Class*. Point estimates range from 0.04 SD to 0.08 SD for general education students, and 0.19 SD to 0.26 SD for students classified both as ELs and with an IEP (see Table

4). In Table 5, we see that relationships between teachers' *Emotional Support* and these two student outcomes consistently are stronger for students identified both as ELs and with an IEP than they are for other students (p = 0.02 to 0.07).

Finally, we find that having a teacher with higher *Classroom Organization* scores results in higher mathematics test scores for general education students and ELs (0.05 to 0.06 SD; see Table 4). Point estimates relating *Classroom Organization* to math test scores of students identified both as an EL and with an IEP are negative in both samples, but not statistically significant. In Table 5, we identify differential relationships between teachers' *Classroom Organization* and students' mathematics test scores with stronger relationships for general education students or ELs relative to students identified as ELs and with an IEP (p = 0.03 to 0.04). Teachers' *Classroom Organization* also predicts general education students' *Self-Efficacy in Math* (0.07 SD; see Table 4), though we do not see differential relationships across student groups (see Table 5). Although teachers' *Classroom Organization* has the strongest theoretical link to students' self-reported *Behavior in Class*, we do not observe strong predictive power here. The only relationship we find between these teacher and student constructs is for general education students in the smaller sample, at the $p < 0.10$ threshold. Further, the magnitude of this relationship (0.04 SD; see Table 4) is not large relative to other point estimates already discussed.

Discussion and Conclusion

Instructional paradigms for mathematics in general education classrooms often describe "good" or "effective" teaching as a messy process in which there are not clear prescriptions for how to do mathematics; rather, students are encouraged to identify and resolve ambiguities through exploration (Brophy, 2002; Cobb & Jackson, 2011; Cohen, 2011; Lampert, 2001).

Instructional reforms that emphasize this sort of ambitious instruction assume that it benefits *all* students in the classroom. However, some advocates for ELs and students with IEPs receiving special education services raise concern that these students may struggle in general education classrooms where the primary focus is on conceptual understanding of mathematics content, at least without additional academic supports and some degree of explicit, direct instruction (Coleman & Goldenberg, 2012; Frizzell, 2013; Schleppegrell, 2007). Because other researchers document the benefits of scaffolded, conceptual-based teaching for students with math-specific learning disabilities (Jitendra et al., 2002; Myers et al., 2015), debates on the relative merits of procedurally versus conceptually based teaching for students with diverse learning needs are ongoing (Agrawal & Morin, 2016).

We find that "good" teaching practices – as conceptualized by reformers, codified in 21st-Century standards, and measured in observation rubrics used both in research and practice settings – have benefits for all types of students when implemented in general education classrooms. Having a teacher who earned high scores on a measure of *Ambitious Mathematics Instruction* positively impacts the mathematics test scores of ELs and students identified both as an EL and with an IEP, as well as the *Behavior in Class* of general education students. This teaching construct also is associated with *Self-Efficacy in Math* for students with an IEP. In fact, we demonstrate that having a teacher who earns high scores on *Ambitious Mathematics Instruction* more strongly benefits the mathematics test scores of ELs than those of general education students. Thus, when planning professional development as well as pre-service training, organizations likely can lean on conceptually demanding instructional techniques in the general education classroom in order to maximize academic, social, and behavioral skills, regardless of the intended student population.

Consistent with theory (Pianta & Hamre, 2009), building relationships and an environment conducive to learning also appears to be important for improving student outcomes. Prior research emphasizes that ELs and students with an IEP, in particular, require social and emotional supports, as well as clear routines and procedures (Niehaus & Adelson, 2014). Indeed, we find that that teachers' interpersonal relationships with students seem to be more strongly associated with student-reported *Self-Efficacy in Math* and *Engagement and Happiness in Class* for students identified both as an EL and with an IEP relative to other students in the general education classroom. Given ongoing pressure that principals and teachers face to raise test scores, education professionals cannot lose sight that teachers also play a vital role in the social and behavioral development of their students. Emphasis on one set of teaching practices (e.g., building positive relationships with students, delivering conceptually demanding mathematics) cannot be a substitute for another.

From a policy perspective, this study disconfirms suspicions we and others had that general education teachers must be overly cautious in deciding which instructional practices to rely on – and, in particular, whether or not to emphasize conceptually demanding mathematics – in order to meet the needs of a diverse group of learners. We did find that the importance of *Ambitious Mathematics Instruction* for different groups of students depended on the outcome examined, suggesting that some adaptations could increase potential benefits. Yet, overall, use of conceptually based teaching shows benefits for all students. Our study also disconfirms a need for specialized tools or protocols to observe the instruction of general education teachers with high concentrations of ELs or students with IEPs who receive special education services. Since we found that measures of "good" or "effective" teaching captured on these instruments generalized across student populations – particularly when we consider a broad range of

outcomes – it seems unnecessary to develop distinct measurement tools or variations of established instruments for general education teachers of students with special needs.

To be clear, we are not suggesting that it is unnecessary for advanced teaching degrees or additional training for teachers who primarily teach ELs or students with an IEP. Further, we are not dismissive of robust policy debates about whether students with specialized academic needs should be taught within or outside of general education classes. Our study observes instruction only in general education classrooms, without making any comparison to instruction delivered in other settings. While IDEA mandates that students receive their schooling in the Least Restrictive Environment (LRE; Individuals with Disabilities Education Improvement Act, 2004), the law also recognizes that students have access to pull-out education services when these supports are better suited for student development. Understanding the relationship between instructional practices and student outcomes in self-contained classrooms or across a combination of settings may be a possibility for future research.

As the student population in U.S. public schools continues to grow more diverse, and as we recognize the unique learning needs that students have, it is important to ensure that policies foster student growth – academically, socially, and behaviorally – for *all* students, and not just the average general education student.

Notes

[1] Instructional delivery focused on conceptual understanding is not mutually exclusive with teaching of correct procedures (Rittle-Johnson & Alibali, 1999). Yet, implementation of cognitively demanding content standards and assessment of their presence in teachers' instruction tend to reward more heavily the former over the latter. For example, one highly used teacher observation instrument, the Framework for Teaching (FFT; Danielson, 2011), provides higher scores for instruction that emphasizes high cognitive challenge and clear explanations of concepts. The two instruments used in this study, the Classroom Assessment Scoring System (CLASS; Pianta, La Paro, & Hamre, 2008) and the Mathematical Quality of Instruction (MQI; Learning Mathematics for Teaching Project, 2011), have a similar emphasis on analysis, problem solving, and reasoning.

[2] Best practices for students with intellectual disabilities include explicit instruction of mathematical thinking (Martinez & Pellegrini, 2010), as well as providing hands-on activities that make complex mathematical processes accessible to students (Göransson, Hellblom-Thibblin, & Axdorph, 2016). Recommendations for teaching students with autism spectrum disorder include utilizing response-prompting strategies or incorporating visual supports into teaching practices (see Knight & Sartini, 2015). And, literature on effective teaching for students with emotional or behavioral disabilities often focuses on teacher-directed opportunities to respond (Ferkis, Belfiore, & Skinner, 1997; Lambert, Cartledge, Heward, & Lo, 2006), especially those teaching practices that incorporate class-wide unison responses (MacSuga-Gage & Simonsen, 2015; Simonsen et al., 2008).

[3] Because student test scores came from administrative data that were mandated for collection by federal law, missingness is quite rare (less than 5% in our sample) and generally not a concern (Card et al., 2010). When constructing analysis files, we dropped students who were missing end-of-year state test scores. The NCTE project also collected data directly from students and teachers. For the student survey, 85% of students who had full test score data completed all survey items; no student skipped the survey altogether. Of the 15% of students who skipped some items, we created composite measures by averaging scores across available items to ensure no missingness. For teacher observations, all teachers in our analysis sample contributed at least one videotaped math lesson.

[4] Teachers were told that videotaped lessons would be used for research purposes only, and they were allowed to schedule recordings during times that were convenient for them and when students were not taking a test. Videos were recorded using a three-camera, unmanned unit; site coordinators turned the camera on prior to the lesson and off at its conclusion. Most lessons lasted between 45 and 60 minutes. Although it is possible that teachers chose their "best" lessons and planned for those, related research projects indicate that teacher rankings are similar when teachers select lessons to be videotaped and when lessons are selected for them (Ho & Kane, 2013).

[5] We excluded four additional items from the *Instructional Support* dimension from the CLASS instrument due to empirical overlap with the items from the *Emotional Support* dimension from

the same instrument (Blazar et al., 2017) and theoretical overlap with many of the items from the MQI.

[6] The intra-class correlation (ICC) calculates the amount of variance in scores attributable to the teacher. Following a generalizability study framework (Hill, Charalambous, & Kraft, 2012), we adjusted ICCs for the median number of lessons per teacher.

[7] For all three outcomes, composite scores that averaged across raw responses were correlated at 0.99 and above with scales that incorporated weights from the factor analysis.

[8] Blazar (2018) showed that teacher effects on the student survey measures used in this study are not particularly sensitive to the inclusion of prior-year measures of the survey versus prior-year test scores. Therefore, we only controlled for prior test scores, which were available for the vast majority of students; prior measures of the survey only were available for a small subset of students who happened to be part of the NCTE project for two years.

[9] We did not use the experimental sample from NCTE in the analyses presented in this paper given a limited sample size of ELs and students with an IEP. Amongst the roughly 1,000 students in the experimental sample, only 7% (N = 78) were identified as having an IEP and 16% (N = 198) as ELs, compared to 11% and 20%, respectively, in the full project sample. These student and teacher sample sizes in the experimental portion of the study do not provide sufficient statistical power to detect differential effects of teaching practice across student subgroups.

[10] It is possible that teaching practices will vary based on student needs in the classroom, in which case researchers may be interested in relating current-year observation scores with current-year student outcomes. However, several analyses to date indicate that characteristics of students in the classroom may lead to bias (Campbell & Ronfeldt, 2018; Steinberg & Garrett, 2016; Whitehurst, Chingos, & Lindquist, 2015), in the same way that value-added scores that estimate the contribution of teachers to student outcomes can be biased when they do not adjust scores for the composition of students in the classroom. Robustness tests, available upon request, that use current-year observation scores or observation scores averaged across all available lessons identify patterns that are similar to those based on use of out-of-year observation scores.

[11] To estimate shrunken observation scores, we specified the following hierarchical linear model separately for each school year:

$$OBSERVATION_{lj,-t} = \gamma_j + \varepsilon_{ljt}$$

The outcome is the observation score for lesson *l* from teacher *j* in years other than *t*; γ_j is a random effect for each teacher, and ε_{ljt} is the residual. For each school year and domain of teaching practice, we utilized standardized estimates of the teacher-level residual as each teacher's observation score in that year. Thus, scores vary across time. In the main text, we refer to these teacher-level residuals as $\widehat{OBSERVATION}_{lj,-t}$ rather than $\hat{\gamma}_j$ for ease of interpretation for readers.

[12] By examining differential teaching effects across subgroups and across samples, we test several bivariate relationships and, in turn, risk seeing false positives. We did not adjust *p*-values

to account for multiple hypothesis testing for two reasons. First, many such tests are seen as far too conservative and so often are set aside for confirmatory results from impact evaluations (Schochet, 2009). Instead, we see many of our results as exploratory due to lingering threats to internal validity that we describe elsewhere in the paper. Second, following Schochet's (2009) recommendation, we took other efforts to reduce the threat of multiple hypothesis testing by focusing on composite measures of teaching practices and composite measures of student outcomes.

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Table 1
Demographic Characteristics of Participating Teachers and Students

	Full Sample	Teachers with Out-of-Year Observation Scores	Teachers with Out-of-Year Observation Scores and Five or More Students with Specialized Academic Needs
Teacher Characteristics			
Male	0.16	0.19	0.20
African-American	0.22	0.17	0.15~
Asian	0.03	0.02	0.02
Hispanic	0.03	0.04	0.04
White	0.65	0.68	0.71
Mathematical Content Knowledge	0.01	0.03	0.03
Alternative Certification	0.08	0.06	0.06
Teaching Experience	10.21	11.74	10.70
Teacher Effects on State Math Test	0.00	0.01	0.01
P-Value on Joint Test for Difference from Full Sample		.504	.643
Teachers	310	196	135
Student Characteristics			
Male	0.50	0.50	0.51*
African-American	0.40	0.38***	0.38**
Asian	0.08	0.09*	0.09**
Hispanic	0.23	0.22*	0.29***
White	0.24	0.27***	0.20***
Eligible for Free or Reduced-Price Lunch	0.64	0.61***	0.73***
English Language Learner (EL) Only	0.17	0.17	0.27***
Has an Individualized Education Plan (IEP) Only	0.08	0.08	0.11***
EL and IEP	0.02	0.02	0.04***
Prior Achievement on State Math Test	0.10	0.14***	-0.05***
P-Value on Joint Test for Difference from Full Sample		.000	.000
Students	10,575	8,660	4,500

~p<0.10, *p<0.05, **p<0.01, *** p<0.001

Table 2
Descriptive Statistics for MQI and CLASS Dimensions

	Univariate Statistics			Pairwise Correlations			
	Mean	SD	Adjusted Intraclass Correlation	Ambitious Mathematics Instruction	Mathematical Errors	Emotional Support	Classroom Organization
Ambitious Mathematics Instruction	1.30	0.13	0.74	1.00			
Mathematical Errors	1.12	0.09	0.56	-0.19**	1.00		
Emotional Support	4.35	0.51	0.53	0.03	-0.12~	1.00	
Classroom Organization	6.39	0.40	0.63	0.08	-0.13~	0.45***	1.00

Notes: Sample includes teachers who were in the study for more than one year and thus have out-of-year observation scores. Intraclass correlations were adjusted for the modal number of lessons. MQI items (from Ambitious Mathematics Instruction and Mathematical Errors) were scored on a scale from 1 to 3. CLASS items (from Emotional Support and Classroom Organization) were scored on a scale from 1 to 7.

~p<.1, **p<.01, ***p<.001

Table 3

Descriptive Statistics for Students' Academic Performance, Attitudes, and Behaviors

	Univariate Statistics			Pairwise Correlations			
	Mean	SD	Internal Consistency Reliability	State Math Test	Behavior in Class	Self-Efficacy in Math	Engagement and Happiness in Class
State Math Test	0.16	0.91	--	1.00			
Behavior in Class	4.15	0.91	0.74	0.24***	1.00		
Self-Efficacy in Math	4.19	0.58	0.76	0.25***	0.35***	1.00	
Engagement and Happiness in Class	4.11	0.85	0.82	0.15***	0.27***	0.62***	1.00

Notes: Sample includes students whose teachers were in the study for more than one year and thus have out-of-year observation scores. For high-stakes math test, reliability varies by district, and so is excluded. Items contributing to Behavior in Class, Self-Efficacy in Math, and Happiness in Class are measured on a 1 to 5 Likert Scale. Statistics were generated from all available data.

*** $p<0.001$

Table 4

Differential Teaching Effects on General Education Students versus those with Specialized Academic Needs

	State Math Test		Self-Efficacy in Math		Engagement and Happiness in Class		Behavior in Class	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
Ambitious Mathematics Instruction								
General Education Students	0.019	0.024	0.006	0.022	0.027	0.063	0.014	0.057*
	(0.015)	(0.024)	(0.020)	(0.029)	(0.036)	(0.059)	(0.018)	(0.024)
English Language Learners (EL) Only	0.055*	0.066*	0.001	0.015	0.027	0.109~	0.024	0.053
	(0.024)	(0.028)	(0.027)	(0.031)	(0.046)	(0.065)	(0.032)	(0.036)
Has an Individualized Education Plan (IEP) Only	-0.011	0.004	0.047	0.110*	0.016	0.027	0.040	0.097*
	(0.028)	(0.036)	(0.046)	(0.055)	(0.054)	(0.070)	(0.040)	(0.047)
EL and IEP	0.085~	0.095*	0.054	0.098~	0.080	0.154	0.041	0.079
	(0.045)	(0.047)	(0.056)	(0.056)	(0.093)	(0.103)	(0.071)	(0.073)
Mathematical Errors								
General Education Students	-0.036*	-0.053**	0.014	0.016	0.030	0.097**	-0.019	-0.001
	(0.016)	(0.020)	(0.021)	(0.024)	(0.033)	(0.036)	(0.015)	(0.019)
EL Only	-0.019	-0.027	0.035	0.025	0.059~	0.085*	0.043~	0.031
	(0.021)	(0.022)	(0.031)	(0.030)	(0.033)	(0.037)	(0.025)	(0.026)
IEP Only	-0.007	-0.002	0.072	0.077	0.091	0.189**	0.075~	0.129*
	(0.030)	(0.033)	(0.049)	(0.059)	(0.065)	(0.066)	(0.041)	(0.051)
EL and IEP	0.039	0.049	-0.040	-0.041	0.017	0.060	-0.043	-0.076
	(0.042)	(0.036)	(0.069)	(0.065)	(0.107)	(0.113)	(0.070)	(0.075)
Emotional Support								
General Education Students	0.012	-0.006	0.042*	0.024	0.081**	0.019	0.009	0.006
	(0.016)	(0.021)	(0.019)	(0.026)	(0.028)	(0.033)	(0.016)	(0.022)
EL Only	0.000	-0.005	0.025	0.033	0.041	0.024	-0.017	-0.031
	(0.024)	(0.026)	(0.028)	(0.031)	(0.037)	(0.045)	(0.025)	(0.028)
IEP Only	0.007	-0.036	0.048	0.037	0.083	0.044	0.032	0.037
	(0.029)	(0.034)	(0.048)	(0.053)	(0.061)	(0.065)	(0.040)	(0.048)
EL and IEP	0.087~	0.079	0.196*	0.188**	0.259**	0.246*	0.080	0.092
	(0.048)	(0.054)	(0.076)	(0.071)	(0.093)	(0.098)	(0.077)	(0.085)
Classroom Organization								
General Education Students	0.001	0.046*	0.010	0.069**	-0.011	0.057	0.027	0.043~
	(0.016)	(0.023)	(0.018)	(0.025)	(0.027)	(0.047)	(0.017)	(0.025)
EL Only	0.056*	0.050~	0.038	0.050	0.010	0.053	0.004	-0.001
	(0.022)	(0.026)	(0.037)	(0.038)	(0.045)	(0.059)	(0.032)	(0.037)
IEP Only	-0.008	0.009	0.043	0.102	0.012	0.044	0.041	0.072
	(0.028)	(0.033)	(0.051)	(0.062)	(0.060)	(0.077)	(0.041)	(0.053)
EL and IEP	-0.044	-0.044	0.011	0.024	-0.070	-0.041	0.047	0.062
	(0.040)	(0.044)	(0.078)	(0.082)	(0.097)	(0.104)	(0.077)	(0.081)
Observations	8,660	4,500	8,660	4,500	5,517	2,515	8,660	4,500
Adjusted R-squared	0.621	0.630	0.068	0.086	0.068	0.092	0.171	0.160

Notes: Estimates in each column come from a single regression model that includes controls for prior student achievement, student demographic characteristics, and class characteristics, as well as school fixed effects. Robust standard errors clustered at the class level in parentheses. For each outcome, model 1 includes the sample of teachers who were in the study for two years

and so have out-of-year observation scores; model 2 further restricts to classes with five or more students with specialized academic needs.

~p<0.10, *p<0.05, **p<0.01, *** p<0.001

Table 5

P-Values on Differences in Teaching Effects between Different Student Groups

	State Math Test		Self-Efficacy in Math		Engagement and Happiness in Class		Behavior in Class	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
Ambitious Mathematics Instruction								
General Education Students versus								
English Language Learners (EL) Only	.084	.060	.868	.809	.990	.277	.736	.884
Has an Individualized Education Plan (IEP) Only	.231	.494	.379	.118	.837	.606	.516	.440
EL and IEP	.134	.097	.414	.193	.571	.347	.716	.788
EL Only versus								
IEP Only	**.042**	.100	.354	.105	.866	.295	.729	.432
EL and IEP	.524	.540	.357	.138	.591	.657	.829	.750
IEP Only versus								
EL and IEP	**.043**	.069	.934	.876	.489	.185	.992	.833
Mathematical Errors								
General Education Students versus								
EL Only	.311	.126	.513	.788	.484	.769	**.038**	.347
IEP Only	.289	.086	.252	.297	.308	.182	**.032**	**.019**
EL and IEP	.067	**.008**	.449	.385	.908	.760	.734	.342
EL Only versus								
IEP Only	.716	.484	.533	.406	.641	.118	.527	.102
EL and IEP	.161	**.047**	.292	.336	.716	.845	.252	.191
IEP Only versus								
EL and IEP	.319	.258	.213	.199	.567	.335	.125	**.016**
Emotional Support								
General Education Students versus								
EL Only	.595	.950	.546	.781	.259	.904	.353	.284
IEP Only	.841	.381	.906	.794	.972	.720	.572	.526
EL and IEP	.116	.119	**.047**	**.029**	.068	**.018**	.350	.312
EL Only versus								
IEP Only	.845	.417	.654	.940	.547	.793	.293	.227
EL and IEP	.089	.145	**.024**	**.033**	**.028**	**.026**	.223	.158
IEP Only versus								
EL and IEP	.111	**.035**	.085	.101	.092	.085	.564	.563
Classroom Organization								
General Education Students versus								
EL Only	**.010**	.865	.455	.614	.644	.947	.504	.305
IEP Only	.758	.328	.546	.624	.704	.879	.762	.604
EL and IEP	.258	**.040**	.992	.586	.549	.349	.806	.815
EL Only versus								
IEP Only	.063	.333	.931	.472	.970	.918	.500	.273
EL and IEP	**.025**	.059	.737	.757	.444	.390	.613	.472
IEP Only versus								
EL and IEP	.403	.257	.709	.419	.401	.422	.943	.914

Notes: P-values less than or equal to 0.05 are bolded. For each outcome, model 1 includes the sample of teachers who were in the study for two years and so have out-of-year observation scores; model 2 further restricts to classes with five or more students with specialized academic needs.

Appendix Table 1

Item Text for Teacher Observation Instruments

Instrument and Domain	Description
Mathematical Quality of Instruction	
Ambitious Mathematics Instruction	
Linking and Connections	Linking and connections of mathematical representations, ideas, and procedures.
Explanations	Explanations that give meaning to ideas, procedures, steps, or solution methods.
Multiple Methods	Multiple procedures or solution methods for a single problem.
Generalizations	Developing generalizations based on multiple examples.
Mathematical Language	Mathematical language is dense and precise and is used fluently and consistently.
Remediation	Remediation of student errors and difficulties addressed in a substantive manner.
Use Student Productions	Responding to student mathematical productions in instruction, such as appropriately identifying mathematical insight in specific student questions, comments, or work; building instruction on student ideas or methods.
Student Explanations	Student explanations that give meaning to ideas, procedures, steps, or solution methods.
Student Mathematical Questioning and Reasoning	Student mathematical questioning and reasoning, such as posing mathematically motivated questions, offering mathematical claims or counterclaims.
Enacted Task Cognitive Activation	Task cognitive demand, such as drawing connections among different representations, concepts, or solution methods; identifying and explaining patterns.
Mathematical Errors and Imprecisions	
Major Errors	Major mathematical errors, such as solving problems incorrectly, defining terms incorrectly, forgetting a key condition in a definition, equating two non-identical mathematical terms.
Language Imprecisions	Imprecision in language or notation, with regard to mathematical symbols and technical or general mathematical language.
Lack of Clarity	Lack of clarity in teachers' launching of tasks or presentation of the content.
Classroom Assessment Scoring System	
Emotional Support	
Positive Climate	Positive climate reflects the emotional connection and relationships among teachers and students, and the warmth, respect, and enjoyment communicated by verbal and non-verbal interactions.
Teacher Sensitivity	Teacher sensitivity reflects the teacher's timely responsiveness to the academic, social/emotional, behavioral, and developmental needs of individual students and the entire class.
Respect for Student Perspectives	Regard for student perspectives captures the degree to which the teacher's interactions with students and classroom activities place an emphasis on students' interests and ideas and encourage student responsibility and autonomy. Also considered is the extent to which content is made useful and relevant to the students.
Classroom Organization	
Negative Climate	Negative climate reflects the overall level of negativity among teachers and students in the class.
Behavior Management	Behavior management encompasses the teacher's use of effective methods to encourage desirable behavior and prevent and redirect misbehavior.
Productivity	Productivity considers how well the teacher manages time and routines so that instructional time is maximized. This dimensions captures to degree to which instructional time is effectively managed and down time is minimized for students.

Appendix Table 2

Item Text for Student Survey

Domains and Item Text
Behavior in Class
My behavior in this class is good.
My behavior in this class sometimes annoys the teacher.
My behavior is a problem for the teacher in this class.
Self-Efficacy in Math
I have pushed myself hard to completely understand math in this class
If I need help with math, I make sure that someone gives me the help I need.
If a math problem is hard to solve, I often give up before I solve it.
Doing homework problems helps me get better at doing math.
In this class, math is too hard.
Even when math is hard, I know I can learn it.
I can do almost all the math in this class if I don't give up.
I'm certain I can master the math skills taught in this class.
When doing work for this math class, focus on learning not time work takes.
I have been able to figure out the most difficult work in this math class.
Engagement and Happiness in Class
This math class is a happy place for me to be.
Being in this math class makes me feel sad or angry.
The things we have done in math this year are interesting.
Because of this teacher, I am learning to love math.
I enjoy math class this year.